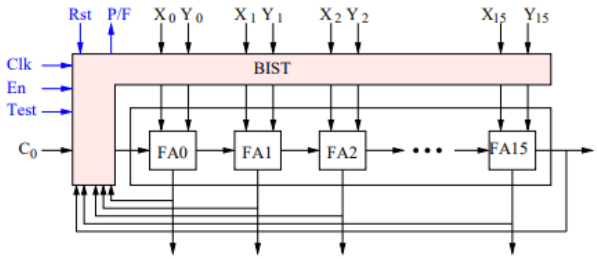
EE6083 VLSI Testing

HW1

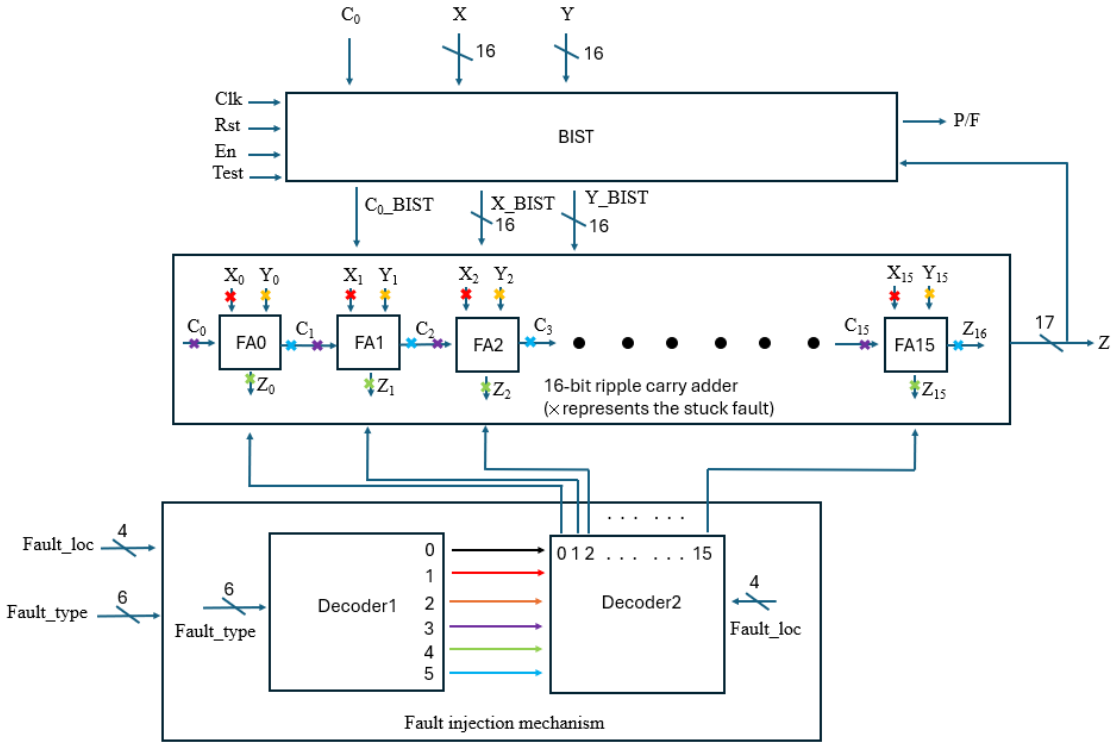
EE6083 VLSI Testing—Homework 1

1. Design a built-in self-test (BIST) circuit for an 16-bit ripple carry adder using Verilog. The BIST should be able to generate test patterns to cover 100% single cell faults. Consider the figure shown below, the additional test pins are in blue. Test is used to switch the circuit between normal mode and test mode. En is used to assert the embedded pattern generator and P/F is used to indicate the ripple carry adder is fault-free or faulty. Verify your BIST design by using fault-injection mechanism. (5 points)



Sol.

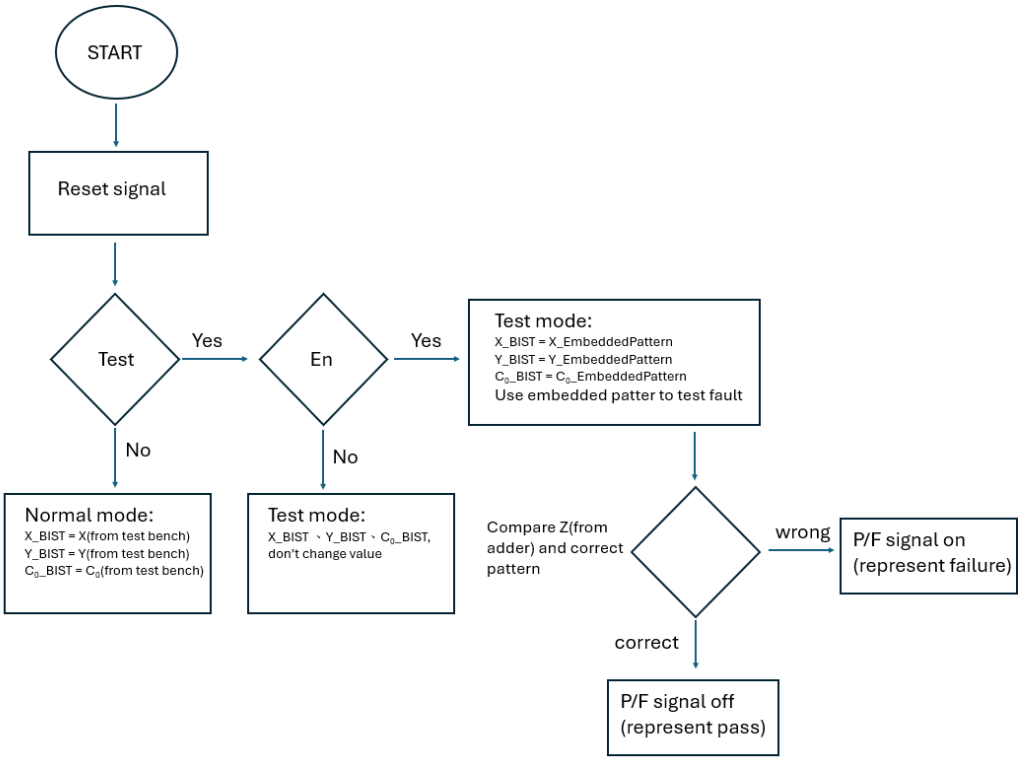
第一小題為設計具有自我測試的 16-bit ripple carry adder 包含 fault injection 和 fault detection，整體的電路架構如圖一。



圖一、16-bit ripple carry adder with BIST 電路架構圖

接下來介紹此電路中相對複雜的部份，分別是 BIST 和 Fault injection mechanism 的實現。

1. BIST: 由於將 Verilog code 貼上來過於多行，我以 flow diagram 的方式呈現 output signal 的邏輯，如圖二。



圖二、BIST logic diagram

大致上的作法為根據 Test 和 En 信號決定輸出信號為何，其中需要特別注意的是做 Fault detection 的 Embedded pattern，對於 Embedded pattern 的決定我們可以根據表一(註:只列出 sum 的部分，Cout 也是一樣的做法)

表一、stuck at fault for sum

Input			Output								
X	Y	Cin	S	$S_{X/0}$	$S_{X/1}$	$S_{Y/0}$	$S_{Y/1}$	$S_{Cin/0}$	$S_{Cin/1}$	$S_{S/0}$	$S_{S/1}$
0	0	0	0	0	1	0	1	0	1	0	1
0	0	1	1	1	0	1	0	0	1	0	1
0	1	0	1	1	0	0	1	1	0	0	1
0	1	1	0	0	1	1	0	1	0	0	1
1	0	0	1	0	1	1	0	1	0	0	1
1	0	1	0	1	0	0	1	1	0	0	1
1	1	0	0	1	0	1	0	0	1	0	1
1	1	1	1	0	1	0	1	0	1	0	1

由於 Embedded pattern 要 cover 100%的 single cell fault，而 single cell fault 共

有 10 種{X/0, X/1, Y/0, Y/1, Cin/0, Cin/1, S/0, S/1, Cout/0, Cout/1}，據觀察發現{0,0,0}和{1,1,1}分別可以 cover {X/1, Y/1, Cin/1, S/1, Cout/1}和{X/0, Y/0, Cin/0, S/0, Cout/0}共十種，所以只需 2 個 pattern 即可。

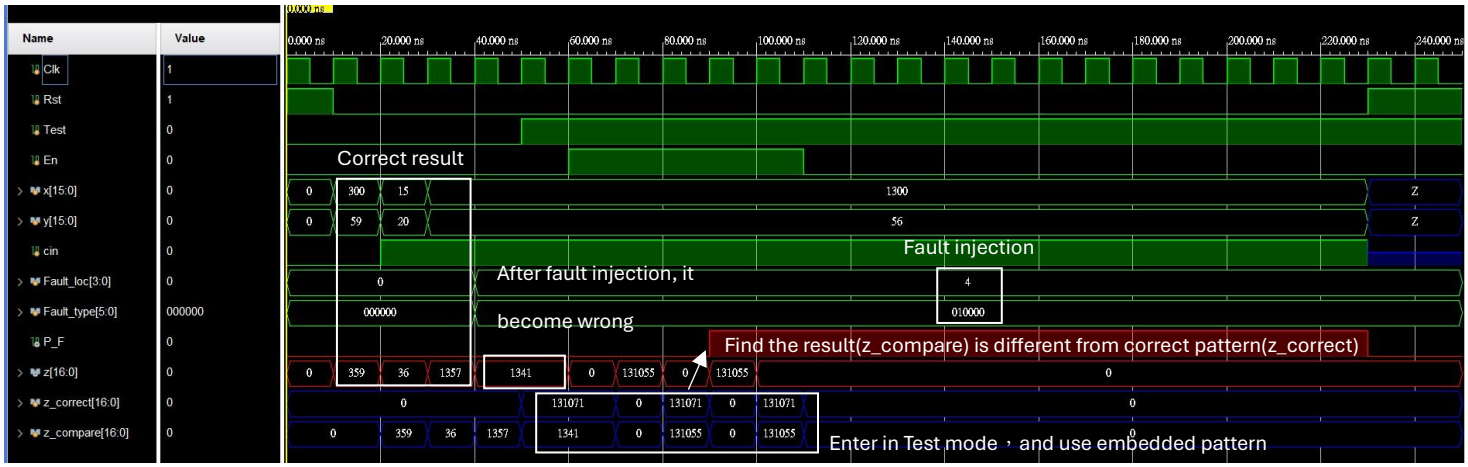
2. Fault injection mechanism

對於 Fault injection 的做法，會由外部輸入兩個訊號 Fault_loc[3:0]和 Fault_type[5:0]，分別決定哪一個 FA 錯誤並且為何種錯，Fault_type[5:0]具體對應何種錯誤可以參考表二(use one hot encoding)

表二、Fault type 的表示

Decoder1	Fault_type[5:1]						
Fault_type[0]		00000	10000	01000	00100	00010	00001
	0	No fault	X is S/0	Y is S/0	Cin is S/0	Z is S/0	Cout is S/0
	1		X is S/1	Y is S/1	Cin is S/1	Z is S/1	Cout is S/1
Output line (will be enable)		0	1	2	3	4	5

波形圖:



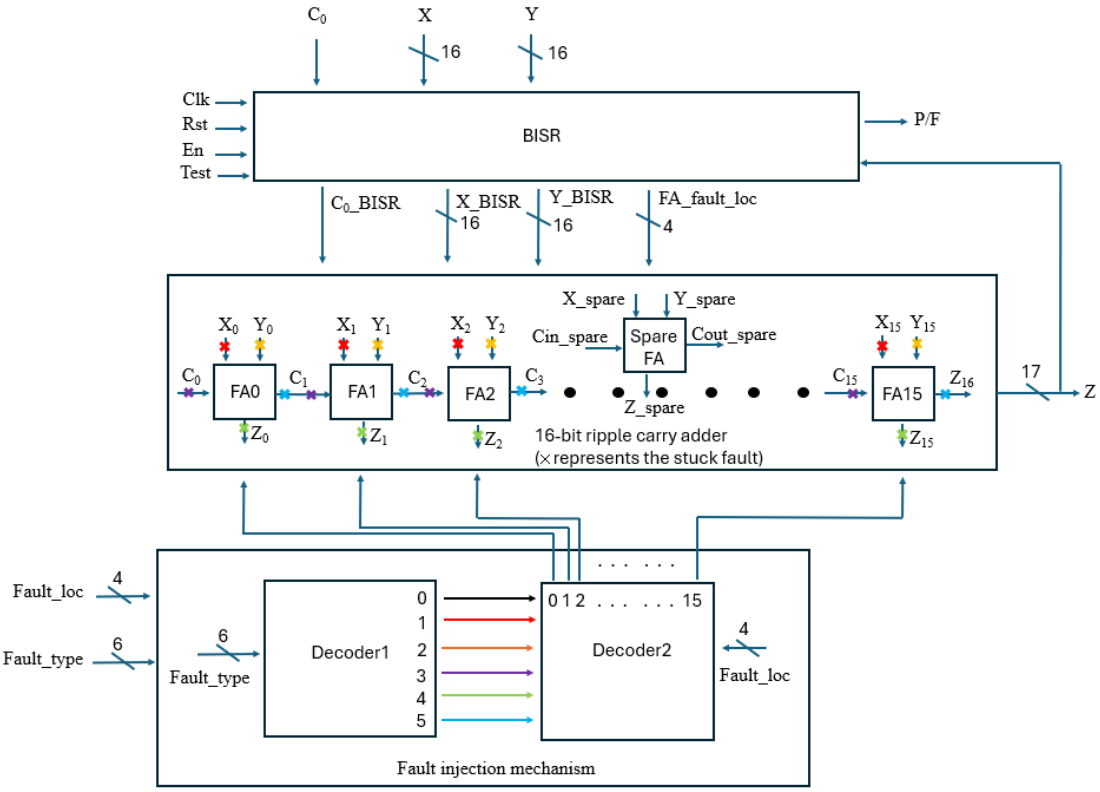
圖三、16-bit ripple carry adder with BIST 波形圖

我們先在 normal mode 進行運作在 fault injection 前，都能正確運算，之後進行 fault injection 後(於 FA4 的 Y stuck at 0)，z[16:0]由原本正確的結果(1357)變為錯誤的結果(1341)，之後於 Test mode 下，也正確做到 fault detection(P_F signal 變為 1)

2. Built-in self-repair (BISR) technique is widely used to improve the yield of regular circuits. Modify the 16-bit ripple carry adder by adding a spare full adder such that it can be repaired if one of the full adders in the ripple carry adder is faulty. Then, design a BISR circuit for the repairable ripple carry adder. Verify your BISR design by using fault-injection mechanism. (5 points)

Sol.

第二小題延續上一題，多了自我修復的功能，利用 spare full adder 將出錯的 FA 替換掉，那首先必須做到 fault location 的判斷



圖四、16-bit ripple carry adder with BISR 電路架構圖

對於 Fault location 判斷其實很簡單因為是 ripple carry 所以我們只要比較回授信號(Z[16:0])與正確的 Pattern(z_correct[16:0])，從 LSB➔MSB 發現第一個不一樣的位數，該位數的位置即為錯誤的 full adder 位置，其 Verilog code 的實現可以參考圖五

```
if(z_compare != z_correct)
    begin
        P_F <= 1'b1;
        casez (z_compare ^ z_correct)
            17'b????_????_????_????_1: FA_fault_loc <= 4'd0;
            17'b????_????_????_???1_0: FA_fault_loc <= 4'd1;
            17'b????_????_????_??10_0: FA_fault_loc <= 4'd2;
            17'b????_????_????_?100_0: FA_fault_loc <= 4'd3;
            17'b????_????_????_1000_0: FA_fault_loc <= 4'd4;
            17'b????_????_???1_0000_0: FA_fault_loc <= 4'd5;
            17'b????_????_??10_0000_0: FA_fault_loc <= 4'd6;
            17'b????_????_?100_0000_0: FA_fault_loc <= 4'd7;
            17'b????_????_1000_0000_0: FA_fault_loc <= 4'd8;
            17'b????_???1_0000_0000_0: FA_fault_loc <= 4'd9;
            17'b????_??10_0000_0000_0: FA_fault_loc <= 4'd10;
            17'b????_?100_0000_0000_0: FA_fault_loc <= 4'd11;
            17'b????_1000_0000_0000_0: FA_fault_loc <= 4'd12;
            17'b???1_0000_0000_0000_0: FA_fault_loc <= 4'd13;
            17'b??10_0000_0000_0000_0: FA_fault_loc <= 4'd14;
            17'b?100_0000_0000_0000_0: FA_fault_loc <= 4'd15;
            17'b1000_0000_0000_0000_0: FA_fault_loc <= 4'd15;
            default: FA_fault_loc <= FA_fault_loc;
        endcase
    end
```

藉由將 z_correct 和 z_compare 做 xor，然後再藉由 priority decoder 得到 fault location

圖五、Fault location 的 Verilog code

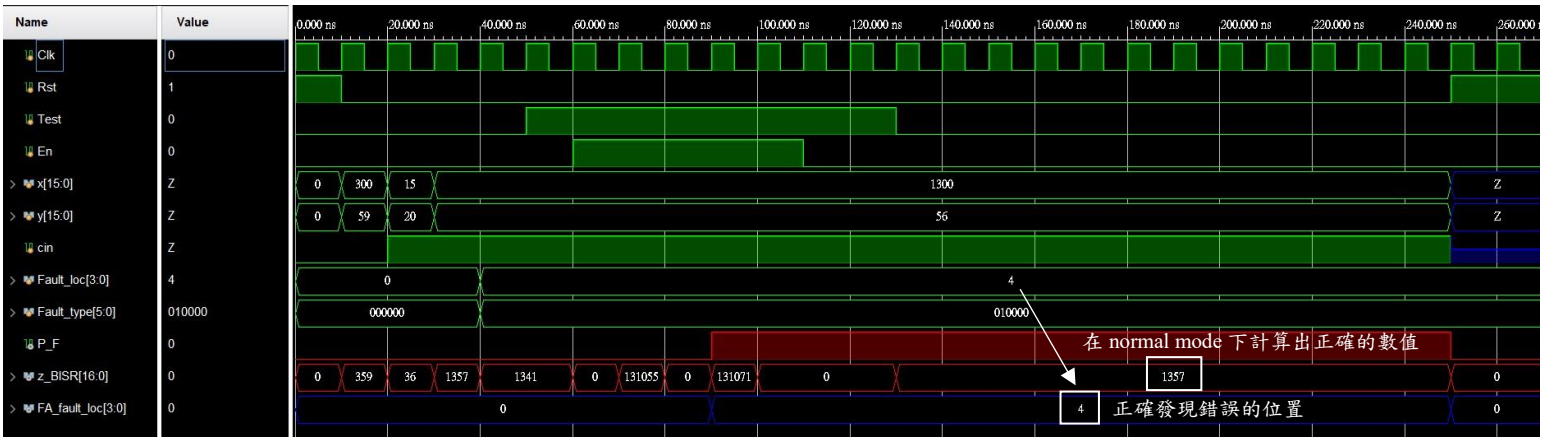
得到 fault location 後，再來就是將錯的 FA 替換掉，其 Verilog code 如圖六

```
if(P_F)  // failure use spare full adder
    begin
        case(FA_fault_loc)
            4'd0:
                begin
                    x_spare = x_BISR[0];
                    y_spare = y_BISR[0];
                    cin_spare = c[0];
                    z_BISR = {z[16:1],z_spare};
                    c_repair = {c[15:2],cout_spare,c[0]};
                end
            4'd1:
                begin
                    x_spare = x_BISR[1];
                    y_spare = y_BISR[1];
                    cin_spare = c[1];
                    z_BISR = {z[16:2],z_spare,z[0]};
                    c_repair = {c[15:3],cout_spare,c[1:0]};
                end
            4'd2:
                begin
                    x_spare = x_BISR[2];
                    y_spare = y_BISR[2];
                    cin_spare = c[2];
                    z_BISR = {z[16:3],z_spare,z[1:0]};
                    c_repair = {c[15:4],cout_spare,c[2:0]};
                end
```

根據 fault location 藉由 decoder 將錯誤的 FA 替換掉

圖六、Replace faulty FA by spare FA 的 Verilog code(只截下部分程式碼)

波形圖:



圖七、16-bit ripple carry adder with BISR 波形圖

圖七的波型，其前半部分與上一小題一致，因此不再多做贅述，後半部分根據 fault location 我們將錯誤的 FA 替換成 spare FA，最後的運算結果可以證實 BISR 電路正確運作。

感謝助教仔細批改!!!